

Frontend Engineering Fundamentals: A 6-Week Intensive Course

WEEK 6: Final Project Completion & Professional Polish

Learning Objectives

By the end of this week, students will:

- Complete the full-stack frontend capstone project
- Implement advanced features and interactions
- Polish UI/UX with animations and transitions
- Write clean, production-ready code
- Deploy and present a professional portfolio project

Core Functionality Completion

Tasks:

- Complete all CRUD operations
- Implement drag-and-drop functionality
- Finalize authentication flow
- Complete all routing and navigation
- Ensure data persistence works flawlessly

Learning Resources:

Videos:

- "DnD kit Tutorial" by Web Dev Simplified (30 mins)
<https://www.youtube.com/watch?v=dL5SOdgMbRY>
- "React Beautiful DnD - Complete Tutorial"
<https://www.youtube.com/watch?v=aYZRRyukuIw>
- "React Performance Optimization" by Fireship (15 mins)
<https://www.youtube.com/watch?v=Qwb-Za6cBws>

UI/UX Polish & Advanced Features

Tasks:

- Implement theme switching (dark/light mode)
- Add animations and transitions
- Create loading skeletons and error boundaries
- Implement search and filter functionality
- Add charts/statistics to dashboard
- Ensure full responsiveness

Learning Resources:

Videos:

- "CSS Animations in React" by Web Dev Simplified (20 mins)
<https://www.youtube.com/watch?v=1wfeqDyMUx4>
- "Dark Mode in React" by Traversy Media (25 mins)
<https://www.youtube.com/watch?v=sy-rRtT84CQ>
- "Framer Motion Tutorial" by Traversy Media
<https://www.youtube.com/watch?v=2V1WK-3HQNk>
- "React Loading Skeletons" by Web Dev Simplified
<https://www.youtube.com/watch?v=ZVug65gW-fc>

Reading:

- Framer Motion Documentation
<https://www.framer.com/motion/>
- Chart.js Documentation
<https://www.chartjs.org/docs/latest/>

- Recharts Documentation
<https://recharts.org/en-US/>

Testing, Optimization & Deployment

Tasks:

- Test all features thoroughly
- Fix bugs and edge cases
- Optimize performance (code splitting, lazy loading)
- Write comprehensive README with screenshots
- Deploy to production
- Set up custom domain (optional)

Learning Resources:

Videos:

- "Deploy React App - Complete Guide" by Traversy Media (30 mins)
https://www.youtube.com/watch?v=Ey_90l9GaAw
- "React Code Splitting" by Web Dev Simplified
https://www.youtube.com/watch?v=JU6sl_yyZqs
- "Lazy Loading in React" by Codevolution
<https://www.youtube.com/watch?v=MJn4W7pR6RU>

Reading:

- React Docs - Code-Splitting
<https://react.dev/reference/react/lazy>
- React Docs - Error Boundaries
<https://react.dev/reference/react/Component#catching-rendering-errors-with-an-error-boundary>

Deployment Checklist:

- All features tested and working
- No console errors or warnings
- Environment variables configured
- Build optimized for production
- README with project description, features, setup instructions
- Screenshots added to README
- Live demo link added
- Repository is public and well-organized

Presentation Preparation

Tasks:

- Create project demo video (2-3 minutes) using loom
- Prepare presentation slides
- Document features and technical decisions

FINAL PROJECT DEMO DAY

Demo Day Structure

Each Student: 10-15 minutes

1. Project Overview

- What problem does it solve?
- Who is the target user?
- Key features highlight

2. Live Demonstration

- Walk through all major features
- Show responsive design
- Demonstrate key interactions
- Highlight technical challenges solved

3. Code Walkthrough

- Show interesting components
- Explain architecture decisions
- Highlight best practices used
- Discuss custom hooks or unique solutions